
What Should a Representation Preserve?

Tayyab Waseem
Department of Computer Science
Lahore University of Management Science
27100171@lums.edu.pk

Abstract

Normal machine learning models rely heavily on the assumption that their data will be IID. However, in real-world scenarios, this assumption falls apart, and regular machine learning models suffer. In this report, I present analysis after conducting experiments on inductive biases, domain adaptation, domain generalisation and open-set recognition.

<https://github.com/TayyabWaseem/beyond-iid>

1 Introduction

Standard supervised learning assumes that training and test data are taken from the same distribution and share the same label space. Whereas in the real world, this is hardly ever true. Objects may appear with different textures, shapes, colors, or belong to entirely unseen classes. This means that a CNN model may achieve strong accuracy on the MNIST dataset, but those same numbers presented in a different setting, such as different colors or backgrounds, might influence the model prediction. However, training models to be strong under these irregularities is very important because in production, they might face a lot of variational data they weren't trained to work with. In my study, I conduct four experiments to answer a few research questions each.

1.1 Inductive Biases & Representations

RQ 1: What do the color and cue-conflict experiments jointly reveal about each model's reliance on shape, texture, and color? How does coverage affect the strength of your conclusions about shape bias?

RQ 2: How do ResNet, ViT, and CLIP respond when spatial structure is altered through translation and patch shuffling? Explain what the results suggest about locality, global organization, and positional sensitivity.

RQ 3: To what extent do prediction-level changes agree with changes in feature representations? Discuss at least one informative agreement or mismatch, and compare CLIP's zero-shot decisions with those of its trained classifier head.

RQ 4: Which observed differences can reasonably be attributed to architecture, and which may instead reflect pretraining data, supervision, augmentation, or model capacity? Use your results and failure cases to justify the distinction.

1.2 Unsupervised Domain Adaptation

RQ 1: How large is the source-to-target domain gap for Source-only ERM, and which classes account for the most important failures?

RQ 2: Across the four methods, does lower domain separability correspond to better target recognition? Use aggregate and class-level evidence to identify successful adaptation or negative transfer.

RQ 3: How does class-conditional alignment compare with the marginal alignment used by DAN and DANN? Do CDAN’s gains or failures support the claim that conditioning helps preserve semantic structure?

RQ 4: How does increasing alignment strength change source performance, domain separability, and target performance? What trade-off does your controlled study reveal, and what setting could have been selected without consulting target labels?

1.3 Domain Generalization

RQ 1: How well do mean-source and worst-source validation performance predict performance on Sketch? Which class-level failures are not visible from the aggregate source results?

RQ 2: Does DAN-DG make the observed source domains less distinguishable, and does that invariance improve Sketch recognition? Identify any evidence that source alignment instead removed class-discriminative information.

RQ 3: Does SAM reduce the common sharpness proxy relative to ERM and DAN-DG? Explain whether the ranking by local stability agrees with the ranking by Sketch performance.

RQ 4: Relative to the shared ERM baseline, what does the comparison between target-aware DAN in Task 2 and target-free DAN-DG in Task 3 suggest about the value of unlabeled Sketch data? What limitations prevent this from being a complete causal attribution?

1.4 Open-Set Recognition

RQ 1: How does semantic similarity affect rejection? Which near and far classes are most often accepted, which CIFAR-10 labels absorb them, and which failures are semantically plausible?

RQ 2: What do MSP, MLS, Energy, and Mahalanobis each capture? Use agreements and disagreements among their scores to explain when confidence, logit magnitude, or feature distance is most informative.

RQ 3: Do GCSC’s random augmentations improve closed-set accuracy, open-set recognition, both, or neither? Explain why its effect differs or remains similar for near and far unknowns.

RQ 4: Does PROSER improve rejection beyond the Vanilla and GCSC models, and what CSA trade-off results? Are its gains consistent with classifier and data placeholders tightening the boundaries between known classes, and where does interpolation between known classes remain insufficient?

2 Methodology & Setup

Tasks 2 and 3 share the same dataset and configuration. However, for the most part, all of the tasks are very distinct from each other.

Task 1

For Task 1, our goal is to answer: How do ResNets, ViTs, and CLIP depend on shape, texture, color, and spatial structure, and does their representation agree with their final prediction? For Task 1, I chose the STL-10 dataset because it has 10 distinct classes (airplane, bird, car, cat, deer, dog, horse, monkey, ship, truck) and a lower computational cost than something like Oxford-IIIT Pets. I used the following frozen pre-trained backbones: ResNet-50 (IMAGENET1K_V2), ViT-B/16 (IMAGENET1K_V1), OpenCLIP ViT-B-32 (pretrained=’openai’).

I froze every backbone and extracted its final representation: the global-average-pooled ResNet feature, the final ViT class token, and the normalized CLIP image embedding. I used the Seed 6304 to create a stratified 80/20 Training/Validation split. I trained classifier heads using the AdamW optimizer with a learning rate of 1e-3, weight decay of 1e-4, for up to 50 epochs with an early stopping patience of 5 epochs. The additional color intervention I chose was Hue Rotation, and cue conflicts were generated via AdaIN. I chose UMAP visualization over t-SNE plots. I selected five maximally contrastive class pairs: (airplane, cat), (car, monkey), (ship, horse), (truck, bird), (dog, airplane); evaluated in both directions (10 directed combinations), generating 20 valid conflict

images per direction (200 total). Images with pixel variance below 100 were rejected as collapsed stylizations. Spatial sensitivity was tested via circular translation (shifts of 0, 8, 16, 32 pixels with reflection padding) and 4x4 patch shuffling (16 patches randomly permuted).

Task 2

In Task 2, I explore Unsupervised Domain Adaptation (UDA). Given labeled source domains and an unlabeled target domain, we ask whether aligning feature distributions can recover lost performance. We specifically compare explicit statistical alignment (DAN) against adversarial alignment (DANN) and class-conditional adversarial alignment (CDAN) to determine if making domains indistinguishable inherently improves target recognition, or if it risks destroying semantic structure.

For this task, I evaluated UDA using the PACS dataset, consisting of Photo, Art Painting, Cartoon, and Sketch (the exact same as the examples given to us in the course). We use Sketch as the unlabeled target. Similar hyperparameters and splits were used as Task 1. I used a **transductive protocol**, in which the target images used for final evaluation are visible during adaptation, but their class labels are not. I utilized a ResNet-18 (IMAGENET1K_V1) backbone with a 7-class linear head. I froze BatchNorm running means and variances at their pretrained ImageNet values across all methods to prevent implicit adaptation via batch statistics and to isolate the effect of the explicit alignment objectives. I trained models using AdamW (LR $1e-4$, WD $1e-4$) for up to 30 epochs, with batches containing 8 examples per source and 24 target examples. As per the experimental design choice, to understand the trade-offs of adversarial alignment, I conducted a controlled study on DANN, sweeping the maximum gradient-reversal strength (max_grl) over 0.25, 0.5, 1.0.

Task 3

In Task 3, we extend this to Domain Generalization (DG), where the target domain is strictly unobservable during training. We evaluate whether explicitly removing source-domain differences (DAN-DG) or optimizing for local parameter stability (Sharpness-Aware Minimization) provides a more reliable route to open-world generalization. For the Domain Generalization experiments, the Sketch domain was strictly hidden during both training and hyperparameter selection.

The models were initialized from the saved Task 2 Source-only ERM checkpoint. I compared the ERM baseline against Sharpness-Aware Minimization (SAM, $\rho=0.05$) and Pairwise Source-Domain Alignment (DAN-DG), which applied Maximum Mean Discrepancy (MMD) across all pairs of the three observed source domains using median-based RBF bandwidths. To analyze the limits of source invariance, I conducted a controlled study varying the DAN-DG alignment strength (lambda_DG) over 0.1, 1.0, 10.0.

Task 4

Finally, in Task 4, we tackle Open-Set Recognition (OSR). We ask how a closed-set classifier can be modified to safely reject unknown classes rather than forcing them into known categories. We evaluate post-hoc novelty scores (MSP, MLS, Energy, Mahalanobis) and compare them against methods that alter the representation itself, specifically strong augmentation (GCSC) and the generation of synthetic classifier and data placeholders (PROSER).

I evaluated OSR using CIFAR-10 as knowns (90/10 split, Seed: 6304) and specific Near and Far classes extracted from the CIFAR-100 test set as unknowns. I adapted a ResNet-18 architecture for 32x32 inputs by replacing the first convolution with a 3x3 stride-1 convolution and removing the initial max-pool. Vanilla and GCSC were trained with SGD (LR 0.1, momentum 0.9, WD $5e-4$, cosine decay) for 100 epochs, with GCSC adding RandAugment(num_ops=2, magnitude=9). PROSER was initialized from the Vanilla checkpoint, appended with 5 dummy classifiers, and trained for 50 epochs using manifold mixup data placeholders and classifier placeholders. An optional extension, Reciprocal Point Learning (RPL) was not conducted, leaving room for future experimentation.

Table 1: Baseline Accuracies

Model	Top-1 Acc.	Macro F1	Mean Max Conf.
ResNet-50 (Linear Head)	0.966	0.966	0.888
ViT-B/16 (Linear Head)	0.968	0.968	0.968
CLIP (Linear Head)	0.964	0.964	0.241
CLIP (Zero-Shot)	0.936	0.936	0.925

3 Results

3.1 Task 1: Inductive Biases

All three backbones achieved strong clean accuracy on STL-10, with ResNet-50 at 96.6%, ViT-B/16 at 96.8%, and CLIP (linear probe) at 96.4%. CLIP’s zero-shot classifier achieved 93.6%, demonstrating strong transfer from language-image pretraining without any task-specific training.

Table 2: Intervention results

Model	Clean	Grayscale		Hue		Shuffle	
	Acc.	Acc.	Cons.	Acc.	Cons.	Acc.	Cons.
ResNet-50	0.966	0.944	0.954	0.942	0.946	0.876	0.898
ViT-B/16	0.968	0.946	0.976	0.928	0.956	0.898	0.916
CLIP (Linear)	0.936	0.908	0.922	0.916	0.930	0.806	0.808
CLIP (Zero-Shot)	0.936	0.908	0.922	0.916	0.930	0.806	0.808

Color removal had a modest impact across all architectures. Grayscale conversion dropped ResNet-50 and ViT by 2.2 points, suggesting that all models rely primarily on luminance patterns rather than chromatic information. Hue rotation produced slightly larger drops for ViT (4.0 points to 92.8%) than ResNet (2.4 points to 94.2%), hinting that ViT may exploit color relationships more than its grayscale robustness would suggest. Consistency metrics (fraction of images where clean and intervened predictions agree) remained high (>92%) for color interventions, telling us that the same images tend to be affected. Patch shuffling produced a lot of divergence. ResNet-50 dropped to 87.6% (9.0 points) while maintaining relatively high prediction consistency (89.8%), indicating that many images were still classified correctly despite spatial destruction. CLIP zero-shot suffered the worst absolute drop (to 80.6%, 12.4 points), with consistency collapsing to 80.8%, suggesting that CLIP’s contrastive text-image alignment is particularly sensitive to spatial coherence.

Table 3: Cue-conflict shape bias

Model	n_shape	n_texture	n_text	Shape Bias (%)	Coverage (%)
ResNet-50	31	36	133	46.3	33.5
ViT-B/16	70	28	102	71.4	49.0
CLIP (Linear)	81	14	105	85.3	47.5
CLIP (Zero-Shot)	81	14	105	85.3	47.5

The cue-conflict experiment (Table 3) exposed the largest architectural difference. ResNet-50 classified based on shape only 31 times versus texture 36 times (shape bias 46.3%). ViT-B/16 strongly favored shape (71.4%), and CLIP was overwhelmingly shape-biased (85.3%).

Representation stability analysis (Table 4) revealed an important nuance. Under patch shuffling, ViT’s feature-level stability was lower than ResNet’s, even though ViT’s prediction accuracy was higher. This dissociation between prediction stability and representation stability suggests that ViT’s features are more globally reorganized by shuffling (because self-attention aggregates across all patches, so permuting patches changes the attention patterns), yet the linear classifier head can still extract the correct class from these reorganized features. ResNet’s features, computed via local convolutions, are less disrupted at the representation level, but the final classifier is less robust to the resulting spatial incoherence.



Figure 1: Example cue-conflict stimuli generated via AdaIN style transfer

Table 4: Representation stability

Model	Grayscale	Shuffle	Translation	Cue-Conflict
ResNet-50	0.787	0.718	0.952	0.213
ViT-B/16	0.737	0.642	0.977	0.177
CLIP	0.894	0.730	0.966	0.651

CLIP exhibited notably high cue-conflict representation stability (0.651) compared to ResNet (0.213) and ViT (0.177). This indicates that CLIP’s feature space preserves similar geometric structure regardless of whether an image is classified by shape or texture and its representations are more style-invariant than the supervised backbones, likely because contrastive language-image pretraining encourages features that match semantic text descriptions e.g. "a photo of a ship" rather than visual statistics.

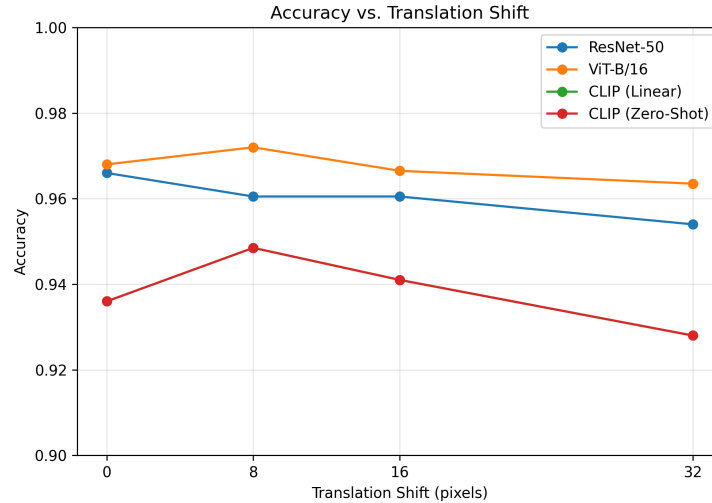


Figure 2: Classification accuracy as a function of translation shift

Under translation, all models maintained high accuracy even at 32-pixel shifts, but ViT was the most stable (consistency 0.987 at shift=32) followed by CLIP (0.964) and ResNet (0.978). The small advantage for ViT is consistent with its global receptive field: since every patch attends to every other patch, moderate spatial displacement has minimal impact on the aggregated [CLS] token.

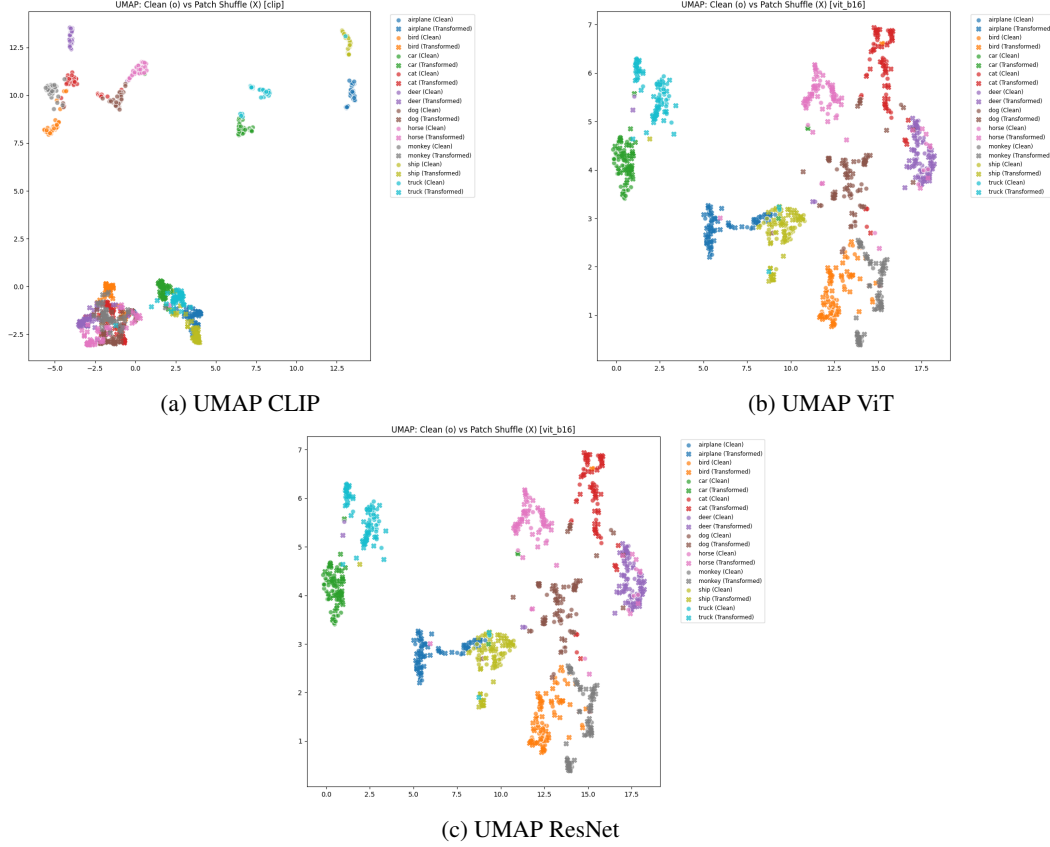


Figure 3: UMAP Visualizations

3.2 Task 2: Unsupervised Domain Adaptation

The Source-only ERM achieved 94.17% mean source accuracy but only 63.76% on Sketch, which is approximately a 30-point domain gap driven by class-specific failures, particularly house (21.3%) and person (40.6%), where Sketch removes the photographic texture cues the model relied on from source domains.

DAN was the only method to achieve positive transfer (69.46%, +5.7 points), reducing domain separability to 84.15% and producing dramatic per-class gains (house: +78.8 points, person: +33.1 points), though giraffe degraded by 19.8 points. Both adversarial methods failed: DANN and CDAN dropped Sketch accuracy to 61.16% and 60.68% respectively while also hurting source performance, and domain separability remained near-ceiling (99.7%), indicating the discriminator was never successfully confused and that the reversed gradients destabilized the backbone without achieving alignment.

Table 5: Task 2 source-domain performance

Method	Photo	Art	Cartoon	Mean Src Acc	Mean Src F1
Source-only	96.74	90.02	95.75	94.17	93.92
DAN	96.74	92.46	95.97	95.05	94.97
DANN	94.96	88.81	86.20	89.99	90.09
CDAN	94.07	82.00	89.60	88.55	88.25

CDAN’s class-level results are interesting. While it improved house (+68.8 points) and horse (+33.9 points), it catastrophically destroyed performance on dog (59.3% $\rightarrow$ 16.7%, 42.6 points) and giraffe (81.8% $\rightarrow$ 24.0%, 57.8 points), which is insane. The confusion matrix shows that 276 of 772 dog

Table 6: Task 2 target-domain performance and domain separability

Method	Target Acc	Target F1	Δ Target	Domain Sep.
Source-only	63.76	57.64	—	99.59
DAN	69.46	63.93	+5.70	84.15
DANN	61.16	64.41	−2.60	99.73
CDAN	60.68	60.57	−3.08	99.59

sketches were misclassified as horse, and 391 of 753 giraffe sketches were misclassified as horse. The class-conditional outer product used by CDAN may have collapsed the feature representations of visually similar animal classes into a single cluster dominated by horse-like features.

Table 7: Per-class target-domain (Sketch) accuracy

Method	Dog	Elephant	Giraffe	Guitar	Horse	House	Person
Source-only	59.33	56.35	81.81	85.86	50.25	21.25	40.63
DAN	57.12	79.05	62.02	76.97	69.85	100.00	73.75
DANN	60.10	47.16	48.47	73.36	69.61	85.00	89.38
CDAN	16.71	85.95	24.04	92.27	84.19	90.00	73.75

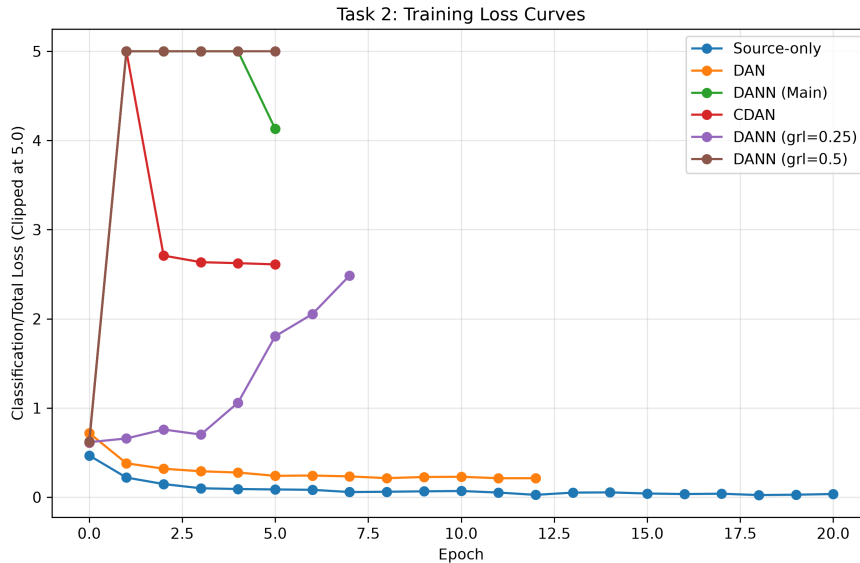


Figure 4: Training losses curve

The training loss curves reveal the mechanism of adversarial failure. Source-only and DAN losses decrease smoothly from 0.47 and 0.72 toward near-zero. In stark contrast, DANN’s total loss explodes to 56,430 at epoch 1, and CDAN’s spikes to 438 at epoch 1. This is a textbook gradient explosion caused by the GRL: when the domain discriminator achieves high confidence early in training, the reversed gradients are massive, and they catastrophically overwrite the pretrained feature representations in a single update. Early stopping at epoch 5-6 salvages a partially functional model (the one we evaluate), but the damage to the feature space is irrecoverable.

The controlled study on DANN’s GRL strength reveals that reducing max_grl from 1.0 to 0.25 partially stabilizes training (the loss spike is smaller) and preserves higher source accuracy (92.21% vs. 89.99%), but fails to improve target accuracy (60.50% vs. 61.16%). All three GRL strengths maintain domain separability above 99.5%, confirming that adversarial alignment fundamentally

Table 8: Effect of the DANN gradient reversal layer (GRL) coefficient

Max GRL	Mean Src Acc	Target Acc	Domain Sep.
0.25	92.21	60.50	99.59
0.50	90.13	61.39	99.86
1.00 (default)	89.99	61.16	99.73

fails on this domain gap regardless of strength. The training logs show that even at $\text{grl}=0.25$, the adversarial loss eventually destabilizes the features, triggering early stopping.

3.3 Task 3: Domain Generalization

Table 9: Task 3 source and target-domain performance

Method	Mean Src Acc	Worst Src Acc	Sketch Acc	Sketch F1	Δ Sketch
ERM	94.17	90.02	63.76	57.64	—
DAN-DG	93.18	88.56	66.38	66.01	+2.62
SAM	95.65	93.43	65.21	68.15	+1.45
DAN-DG ($\lambda=0.1$)	95.82	93.43	69.94	71.31	+6.18
DAN-DG ($\lambda=10.0$)	21.64	17.20	4.07	1.12	-59.68

Table 10: Task 3 source-domain separability and loss-landscape sharpness at the found minimum.

Method	Source Sep.	Sharpness
ERM	84.21	0.825
DAN-DG	61.84	0.890
SAM	83.22	0.334
DAN-DG ($\lambda=0.1$)	69.08	0.483
DAN-DG ($\lambda=10.0$)	33.22	0.133

All Task 3 methods were initialized from the identical Task 2 ERM checkpoint (94.17% mean source, 63.76% Sketch accuracy), ensuring a controlled comparison.

SAM improved Sketch accuracy to 65.21% (+1.5 points) and, notably, produced the highest mean source accuracy (95.65%) and worst-source accuracy (93.43%, Art Painting) among all Task 3 methods. This mean that SAM does not sacrifice source performance to improve generalization but instead by seeking flat minima, it simultaneously improves performance on the observed domains and the unseen target. The sharpness proxy dropped dramatically from 0.825 (ERM) to 0.334, confirming that SAM successfully found a substantially flatter region of the loss landscape. Importantly, source domain separability remained high (83.22%, versus ERM’s 84.21%), proving that SAM’s generalization mechanism is fundamentally different from domain alignment.

DAN-DG at $\lambda=1.0$ improved Sketch accuracy to 66.38% (+2.6 points) while reducing source separability to 61.84% (from 84.21%), directly confirming that MMD alignment across the three source domains successfully removed domain-identifying information from the features. However, DAN-DG’s sharpness (0.890) was actually higher than ERM’s (0.825), showing that alignment and flatness are independent mechanisms that affect generalization through different pathways.

Per-class analysis reveals complementary strengths. SAM dramatically improved dog recognition (+24.9 points) and house (+50.0 points), but severely degraded elephant (26.9 points). DAN-DG excelled on person (+53.1 points) and horse (+22.4 points), but suffered on giraffe (-31.7 points) and dog (5.3 points). Neither method uniformly dominates the other, suggesting that the optimal generalization strategy may be class-dependent.

The controlled study on DAN-DG alignment strength reveals a dramatic non-linear trade-off. At $\lambda=0.1$, the model achieved the best Sketch accuracy across all Task 2 and Task 3 methods: 69.94% (+6.2 points over ERM), with improved source accuracy (95.82%), moderate separability reduction

Table 11: Per-class target-domain (Sketch) accuracy for ERM, DAN-DG, and SAM.

Method	Dog	Elephant	Giraffe	Guitar	Horse	House	Person
ERM	59.33	56.35	81.81	85.86	50.25	21.25	40.63
DAN-DG	54.02	62.43	50.07	92.76	72.67	56.25	93.75
SAM	84.20	29.46	69.32	94.41	52.33	71.25	71.25

Table 12: Controlled study: effect of the domain-adversarial loss weight (λ_{DG}) on source accuracy, target (Sketch) accuracy, source-domain separability, and loss-landscape sharpness.

λ_{DG}	Mean Src Acc	Sketch Acc	Source Sep.	Sharpness
0.1	95.82	69.94	69.08	0.483
1.0 (default)	93.18	66.38	61.84	0.890
10.0	21.64	4.07	33.22	0.133

(69.08%), and improved sharpness (0.483). This gentle alignment provided the optimal balance between removing domain-specific noise and preserving class-discriminative structure.

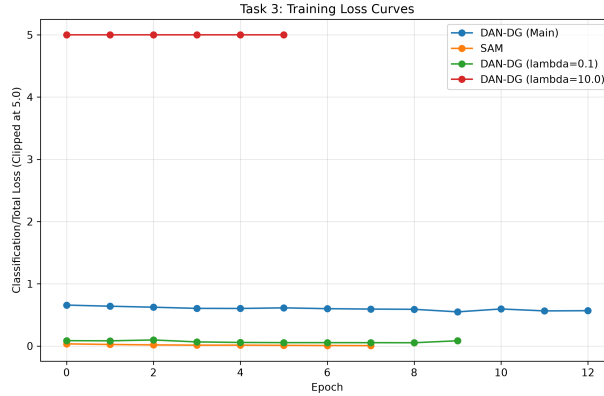


Figure 5: Training losses curve

At $\lambda=10.0$, complete catastrophic collapse occurred. Mean source accuracy crashed to 21.64%, and Sketch accuracy to just 4.07%. The per-class breakdown reveals the mechanism: the model predicted "person" for 100% of all inputs, achieving 100% accuracy on person and 0% on all six other classes. Source separability dropped to 33.22% confirming that the features were indeed domain-invariant, but trivially so: by mapping every input to the same point in feature space, the model achieved perfect invariance at the cost of zero discrimination. The sharpness proxy at $\lambda=10.0$ (0.133) was even lower than SAM's (0.334), but this flatness is meaningless as the loss surface is flat because the model has converged to a degenerate constant-output solution.

3.4 Task 4: Open-Set Recognition

The Vanilla model achieved a closed-set accuracy (CSA) of 94.69% on CIFAR-10 test data. When applying post-hoc scoring rules to the frozen Vanilla logits, a consistent pattern emerged: Far unknowns are substantially easier to reject than Near unknowns across all scoring methods. In the feature space, Far unknowns (bottle, chair, wardrobe) lie in regions geometrically distant from all class centroids, making distance-based rejection straightforward. Near unknowns (wolf, leopard, fox) map close to the dog/cat centroids, making them nearly indistinguishable from known classes in feature space.

Among the logit-based scores, MSP achieved the highest overall AUROC (85.89%) by transforming logits through softmax, which compresses high-confidence predictions and amplifies uncertainty.

Table 13: Post-hoc OOD detection scores on the Vanilla backbone

Score	CSA	AUROC Near	AUROC Far	AUROC All
MSP	94.69	82.45	89.33	85.89
MLS	94.69	81.00	88.48	84.74
Energy	94.69	81.02	88.52	84.77
Mahalanobis	94.69	73.57	91.11	82.34

Table 14: Post-hoc OOD detection scores on the Vanilla backbone

Score	FPR@95 Near	FPR@95 Far
MSP	70.25	56.75
MLS	64.88	46.50
Energy	64.00	45.13
Mahalanobis	84.88	49.13

MLS (84.74%) and Energy (84.77%) performed very similarly, which is expected since Energy (logsumexp) and MLS (max) differ only in how they aggregate the logit vector, and for peaked distributions, logsumexp $\approx$ max. However, at the operating threshold (FPR@95), MLS and Energy substantially outperformed MSP because the raw logit magnitude provides better calibrated separation than the squashed softmax probability. Mahalanobis distance presented an interesting failure mode. Despite achieving the highest Far AUROC (91.11%), it had the worst Near FPR@95 (84.88%), meaning that at the 95% TPR threshold, nearly 85% of Near unknowns were incorrectly accepted.

Table 15: Comparison of OOD detection performance pt. 1

Method (Score)	CSA	AUROC Near	AUROC Far	AUROC All
Vanilla (MLS)	94.69	81.00	88.48	84.74
GCSC (MLS)	95.34	82.12	91.34	86.73
PROSER (MLS)	94.61	69.22	49.40	59.31
PROSER (Placeholder)	94.61	77.08	91.76	84.42

CSC (RandAugment) improved both closed-set accuracy and open-set rejection across the board. By training on heavily augmented images, GCSC forces the model to classify based on essential semantic features rather than superficial pixel patterns, producing a more compact and well-separated feature space that naturally separates unknowns. PROSER’s results require careful interpretation. When evaluated with standard MLS scoring, PROSER performed catastrophically (59.31% overall AUROC), because the 5 dummy logits shift the overall logit distribution. However, PROSER’s dedicated placeholder score achieved 77.08% Near AUROC and 91.76% Far AUROC, which is the highest Far AUROC across all methods. It also achieved the lowest Far FPR@95, substantially outperforming Vanilla MLS and GCSC MLS. This demonstrates that the manifold mixup data placeholders successfully carved out "unknown territory" in the inter-class boundary regions, and the dummy classifiers learned to activate for inputs that fall in these regions.

However, PROSER’s Near AUROC (77.08%) was lower than both Vanilla MSP (82.45%) and GCSC MLS (82.12%). This suggests that interpolation between known-class features at the layer2/layer3 boundary does not adequately simulate Near unknowns, which are not feature-space interpolations of known classes but rather genuinely novel visual categories that happen to share surface-level features.

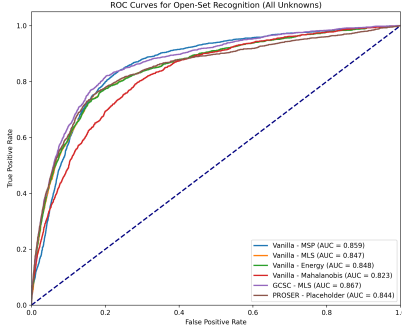
4 Discussion

4.1 Visual Cues and Distribution Shift

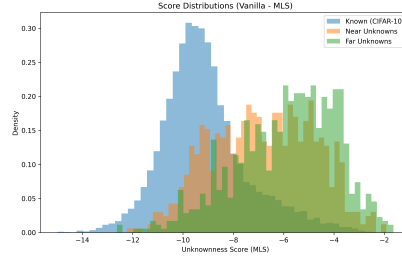
Task 1 established that ResNet-50 relies heavily on local texture (shape bias 46.3%), while ViT-B/16 (71.4%) and CLIP (85.3%) are progressively more shape-biased. The PACS Sketch domain is defined precisely by the removal of photographic texture, preserving only structural outlines without color,

Table 16: Comparison of OOD detection performance pt. 2

Method (Score)	FPR@95 Near	FPR@95 Far
Vanilla (MLS)	64.88	46.50
GCSC (MLS)	66.88	42.88
PROSER (MLS)	82.25	93.75
PROSER (Placeholder)	70.00	39.13



(a) ROC Curves



(b) Score Distributions MLS

Figure 6

shading, or material appearance. Tasks 2 and 3 used a ResNet-18 backbone, which shares the same convolutional, locally-receptive architectural family as ResNet-50, and this backbone suffered a 30-point accuracy drop from source (94.2%) to Sketch (63.8%).

This qualitative correspondence is suggestive: the CNN backbone’s reliance on local statistical patterns (textures) rather than global shape is a plausible contributing factor to the Sketch domain gap. When texture is stripped away (as it is by definition in Sketch) the features that the network has learned to rely on become uninformative, and classification performance collapses. The per-class breakdown reinforces this interpretation. The two classes with the lowest Source-only Sketch accuracy, house and person, are precisely the classes where photographic texture carries the most discriminative weight in the source domains, as in, brick and roofing patterns for houses, skin and clothing texture for people. In Sketch, these cues vanish entirely, leaving only geometric outlines that the texture-biased CNN has not learned to exploit.

However, this link must be treated as qualitative rather than causal. Task 1 evaluated ResNet-50 on STL-10, while Tasks 2 and 3 evaluated ResNet-18 on PACS. We cannot rule out confounds arising from these differences, nor can we claim that replacing the CNN backbone with a shape-biased ViT would necessarily close the Sketch gap, since domain shift involves more than just texture removal.

4.2 Invariance vs. Discriminability

The results across Tasks 2 and 3 reveal a consistent pattern: moderate domain invariance improves generalization, but aggressive invariance destroys class discrimination. Two distinct regimes emerge. In the moderate invariance regime, DAN (separability 84.2%) and DAN-DG at $\lambda=0.1$ (69.1%) both achieved substantial target accuracy improvements by gently blending domain-specific features without erasing the class-discriminative boundaries between them. Statistical alignment via MMD acts as a soft regularizer that nudges representations toward a shared subspace while preserving semantic structure.

In the adversarial failure regime, DANN and CDAN failed to reduce separability at all (both >99%), yet still caused negative transfer. The training loss curves reveal why: the GRL-reversed gradients caused catastrophic parameter updates in the first epoch, destroying the pretrained feature representations without achieving any actual alignment. The domain discriminator was never fooled; the features were simply corrupted.

At the opposite extreme, DAN-DG at $\lambda=10$ achieved near-perfect invariance (separability 33.2%), but at the cost of total discrimination collapse: the model predicted “person” for 100% of inputs. This confirms that domain invariance is not inherently beneficial and it is only useful when it removes domain-specific noise without erasing the class-conditional structure.

SAM provides a revealing counterpoint. It achieved strong generalization (65.2% Sketch accuracy) while maintaining *high* source separability (83.2%). SAM’s mechanism is fundamentally orthogonal to alignment: rather than making domains indistinguishable, it finds parameters where the loss surface is flat, so that the implicit “perturbation” of switching domains causes minimal degradation.

4.3 Importance of Target Domain Access

The shared ERM baseline provides a controlled comparison point:

[nosep]

- Task 2 DAN (target-aware, $\lambda=1.0$): 69.46% Sketch accuracy
- Task 3 DAN-DG (target-free, $\lambda=0.1$): 69.94% Sketch accuracy
- Task 3 DAN-DG (target-free, $\lambda=1.0$): 66.38% Sketch accuracy

At comparable alignment strengths ($\lambda=1.0$), target-aware DAN outperforms target-free DAN-DG by 3.1 points, suggesting that explicit access to unlabeled target data provides meaningful directional guidance for feature alignment.

However, DAN-DG at $\lambda=0.1$ slightly exceeds DAN at $\lambda=1.0$ (69.94% vs. 69.46%), complicating this interpretation. The gentler alignment appears to remove just enough domain-specific variance from the source representations to improve generalization, without the risk of overshooting into feature collapse. A fundamental confound prevents clean causal attribution: DAN aligns source features to target features, while DAN-DG aligns source features to each other, and the two methods were evaluated at different optimal λ values.

4.4 Recognition vs. Rejection

Task 4 connects directly to the earlier tasks’ theme of models being “confident but wrong.” In Tasks 2 and 3, the ERM model confidently classified Sketch images incorrectly because it had encoded texture-based features that are absent in edge-map drawings. In Task 4, the same phenomenon manifests differently: the Vanilla classifier confidently assigns wolf images to the “dog” class because it has learned dog-like features without understanding where the boundaries of “dogness” end in feature space.

GCSC and PROSER represent two distinct approaches. GCSC smooths the decision boundary by training on heavily augmented images, forcing the classifier to rely on essential semantic features. This produces a more compact known-class volume whose boundaries naturally exclude more unknowns, improving closed-set accuracy and open-set rejection simultaneously (analogous to how DAN gently blends distributions). PROSER takes a more aggressive approach, explicitly carving out “unknown territory” in the feature space by generating synthetic boundary samples via manifold mixup. This proves highly effective for Far unknowns (AUROC 91.76%), because Far unknowns lie in regions of feature space that *can* be interpolated between known classes. However, PROSER underperforms GCSC on Near unknowns (77.1% vs. 82.1% AUROC), because Near unknowns are not feature-space interpolations of known classes e.g. a wolf is not “halfway between” a dog and a cat in any meaningful representational sense; it is a genuinely novel visual category. This parallels how DANN’s aggressive adversarial alignment in Task 2 disrupted features without achieving meaningful adaptation: more aggressive interventions do not always yield better outcomes.

4.5 Overall Synthesis

To answer the title of my report “What Should a Representation Preserve?”, a coherent picture emerges of what a robust representation must *preserve*, *suppress*, and *reject*.

Preserve. A robust representation must encode shape-based, semantic features that transfer across visual domains. Task 1 showed that shape-biased architectures are inherently more robust, and Tasks

2 and 3 confirmed that the texture-biased CNN backbone suffers precisely when texture is removed. Equally important, the representation must preserve class-discriminative boundaries: DAN in Task 2 and GCSC in Task 4 both improved target-domain or open-set performance *while maintaining or improving* source/closed-set accuracy.

Suppress. Domain-specific texture and style information should be suppressed, but *gently*. DAN’s moderate MMD alignment and DAN-DG at $\lambda=0.1$ successfully removed domain-identifying features while preserving semantic structure. In contrast, aggressive suppression consistently backfired: DANN’s adversarial gradients destabilized features, CDAN collapsed visually similar classes, and DAN-DG at $\lambda=10$ erased class information entirely. SAM’s flat minima improved generalization through an entirely orthogonal mechanism to domain alignment, showing that Loss landscape sharpness should also be suppressed.

Reject. Inputs that fall outside the known-class feature volume should be identified and rejected rather than forcibly classified. Task 4 showed that PROSER’s placeholder scoring effectively rejects Far unknowns by explicitly bounding the known-class volume. However, the fundamental difficulty revealed is that Near unknowns cannot be reliably rejected by any approach tested. This suggests that achieving robust open-set recognition for semantically similar unknowns may require a deeper representational change.

5 Conclusion

This report systematically evaluated model robustness across four paradigms of increasing difficulty. Task 1 demonstrated that CNNs rely on local texture with a shape bias of 46.3%, while Vision Transformers favor global shape (71.4%, 85.3%). Task 2 showed that statistical domain alignment (DAN) successfully bridges the source-to-Sketch gap (+5.7 points), while adversarial methods suffer gradient explosion and negative transfer. Task 3 proved that generalization to unseen domains is achievable without target data: SAM reduces landscape sharpness and improves Sketch accuracy, while gentle source alignment (DAN-DG $\lambda = 0.1$) achieves the best overall Sketch accuracy of 69.9%. Task 4 demonstrated that strong augmentation (GCSC) and synthetic placeholders (PROSER) improve open-set rejection, with PROSER achieving the best Far FPR@95 of 39.1%.

Limitations & Future Work. I aimed that this research answered all the RQs listed earlier. My experiments used relatively small backbones at modest resolutions, and the cross-task comparisons (e.g., DAN vs. DAN-DG) involve confounds that prevent strict causal attribution. Extending the studies to multiple seeds and testing ViT backbones on downstream tasks would strengthen these findings. Additionally, explicitly modeling class-conditional open-space (e.g., via Reciprocal Point Learning) could improve Near unknown rejection, the weakest link across all evaluated methods.

References

- Foret, P., Kleiner, A., Mobahi, H., & Neyshabur, B. (2021). Sharpness-Aware Minimization for Efficiently Improving Generalization. *International Conference on Learning Representations (ICLR)*.
- Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., & Lempitsky, V. (2016). Domain-Adversarial Training of Neural Networks. *Journal of Machine Learning Research*, 17(59), 1–35.
- Geirhos, R., Rubisch, P., Michaelis, C., Bethge, M., Wichmann, F. A., & Brendel, W. (2019). ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. *International Conference on Learning Representations (ICLR)*.
- Long, M., Cao, Z., Wang, J., & Jordan, M. I. (2018). Conditional Adversarial Domain Adaptation. *Advances in Neural Information Processing Systems (NeurIPS)*.
- Vaze, S., Han, K., Vedaldi, A., & Zisserman, A. (2022). Open-Set Recognition: a Good Closed-Set Classifier is All You Need? *International Conference on Learning Representations (ICLR)*.
- Zhou, D.-W., Ye, H.-J., & Zhan, D.-C. (2021). Learning Placeholders for Open-Set Recognition. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.